

From the Data Package $\mathcal{Q}$ to a Gauge-Fixed Path Integral for the Trajectory-Induced $SU(3)$ Lattice Theory

Abstract

The companion lattice paper constructs from discrete particle trajectories a finite $SU(3)$ lattice gauge theory data package

$$\mathcal{Q} = (\mathcal{N}, \mathcal{E}, \mathcal{C}_{\square}, \{U_3(e)\}_{e \in \mathcal{E}}, S_W, S_F, S_\phi)$$

with explicit worldline links, interaction links, Wilson plaquettes, and matter actions [1, 2]. The present paper performs the next mathematical step: it replaces the data-induced background links by independent integration variables and builds the exact finite-dimensional Euclidean path integral attached to the same lattice. We define the full configuration space, the product Haar–Berezin–Lebesgue measure, the partition function, and the vacuum expectation map. We prove that the resulting finite-lattice theory is well defined whenever the scalar local potential is bounded below. We then solve the gauge-redundancy problem exactly by a rooted maximal-tree gauge: every gauge orbit meets the gauge slice once and only once, the Faddeev–Popov determinant is identically 1, and the ghost sector is a decoupled finite Berezin integral. The fermionic sector integrates to a finite fermion determinant, and the scalar sector is shown to be integrable with an exact Gaussian formula in the free case.

The paper also explains the continuum meaning of the discrete measure. On each edge, Haar measure is shown to be smooth in exponential coordinates and to agree to second order with Lebesgue measure on the Lie algebra. Combined with the continuum Wilson-action limit from the companion lattice paper, this yields a precise consistency statement: the discrete theory is a genuine ultraviolet regularization whose near-identity local measure and local action density match the formal Euclidean Yang–Mills path-integral rules. What is *not* proved here is just as important: this paper does not establish an unconditional constructive continuum 4-dimensional Yang–Mills measure, does not verify the Osterwalder–Schrader axioms in the continuum, and does not prove a mass gap. Those are strictly stronger problems than the finite-lattice quantization solved in this paper.

Contents

1	Introduction and motivation	2
2	Configuration space and measure	3
2.1	Imported lattice data	3
2.2	Field manifold	4
2.3	Measures	5
3	Construction of the partition function	6
4	Gauge redundancy and orbit analysis	7
5	Gauge fixing and the Faddeev–Popov procedure	8

6	Ghost fields and the effective action	11
7	Matter sector integration	13
8	Recovery of the Feynman path-integral measure	16
9	Discussion of excitations and propagators	18
10	Conclusion	19

1 Introduction and motivation

From data to dynamics. The paper *Finite $SU(3)$ Lattice Gauge Theory from Particle Trajectories* constructs from the recorded run data of the viscous particle system a finite spacetime lattice

$$\mathcal{L} = (\mathcal{N}, \mathcal{E})$$

whose site set $\mathcal{N}$ consists of trajectory records $n_{i,t}$, whose edge set $\mathcal{E}$ consists of temporal worldline links and same-time interaction links, and whose canonical plaquette family $\mathcal{C}_\square$ is built from persistent interaction pairs across adjacent times [2]. On that lattice the same paper defines explicit link transporters $U_3(e) \in SU(3)$, the Wilson action S_W , and matter actions S_F and S_ϕ . These objects form the finite data package $\mathcal{Q}$. Up to that stage, however, the link variables are still deterministic functions of the particle record. A quantum theory requires a different mathematical object: instead of one distinguished configuration, it requires an ensemble weighted by an action and summed over all admissible gauge fields and matter fields. The present paper performs that passage from one explicit background configuration to a finite-dimensional path integral on the same lattice.

The quantization objective. The basic quantization step is conceptually simple and mathematically precise. The background transporters $U_3(e)$ derived in [2] are replaced by independent variables

$$U(e) \in SU(3), \quad e \in \mathcal{E},$$

and the site colors $\mathbf{c}_x$ are retained, when desired, as auxiliary fluctuating variables in the unit ball of $\mathbb{C}^3$. Because the companion paper derives no autonomous color-field energy beyond the induced links, the auxiliary color sector will be normalized and dynamically inert in the quantization performed here. All dynamical content therefore sits in the gauge links, fermions, and scalars. The goal is to define a mathematically complete finite partition function, remove gauge redundancy exactly, identify the fermion determinant and scalar measure, and clarify in what sense the resulting finite theory does and does not recover continuum Euclidean Yang–Mills quantization.

Theorem 1.1 (Main finite-lattice quantization result). *Fix the lattice data $(\mathcal{N}, \mathcal{E}, \mathcal{C}_\square)$ from [2], the coupling $g_3 > 0$, masses $m_F > 0$ and $m_\phi > 0$, the distance regularization $\eta > 0$, and a continuous scalar local potential $V_\phi : \mathbb{R} \rightarrow \mathbb{R}$ satisfying*

$$V_\phi(s) \geq -C_\phi \quad \text{for all } s \in \mathbb{R}$$

for some constant $C_\phi < \infty$. Then:

1. *the promoted field manifold*

$$\mathcal{X}_{\mathcal{L}} := SU(3)^{\mathcal{E}} \times \mathbb{B}^{\mathcal{N}} \times \Lambda_F \times \mathbb{R}^{\mathcal{N}}$$

admits a canonical product measure obtained from normalized Haar measure on $SU(3)$, normalized Lebesgue measure on the color ball $\mathbb{B} \subset \mathbb{C}^3$, finite Berezin measure in the fermionic directions, and Lebesgue measure in the scalar directions;

2. the partition function

$$Z = \int_{\mathcal{X}_{\mathcal{L}}} \exp(-S_W(U) - S_F(U, \bar{\psi}, \psi) - S_\phi(\phi)) \, d\mu$$

exists as a finite complex number;

3. for any rooted spanning forest $\mathcal{T}$ of the underlying undirected graph, every rooted gauge orbit meets the maximal-tree gauge slice exactly once, the Faddeev–Popov determinant is identically 1, and the gauge-fixed partition function is an ordinary finite-dimensional integral over the cotree links;
4. integrating out the fermions yields the exact fermion determinant $\det M_F(U)$, while the scalar integral is finite and is Gaussian with covariance $(L_\eta + m_\phi^2 I)^{-1}$ when $V_\phi \equiv 0$.

Remark 1.2 (What this paper does not prove). Theorem 1.1 is a complete quantization theorem for a *finite lattice* and nothing less, but also nothing more. It does not produce an unconditional constructive continuum Yang–Mills measure in four dimensions. In particular, it does not resolve the Yang–Mills Millennium Prize problem, which additionally requires a nontrivial continuum quantum field theory satisfying the Euclidean or Wightman axioms and exhibiting a positive mass gap. The finite-lattice theory constructed here is the rigorous ultraviolet regularization on which such harder questions would have to be based.

2 Configuration space and measure

2.1 Imported lattice data

We summarize only the pieces of the companion paper [2] that are required for quantization.

Definition 2.1 (Imported lattice package). *The previous paper supplies:*

1. a finite site set $\mathcal{N} = \{n_{i,t}\}$ and a finite oriented edge set $\mathcal{E} = E_{\text{time}} \cup E_{\text{int}}$;
2. for every plaquette $P \in \mathcal{C}_\square$, an oriented boundary

$$\partial P = (e_1, e_2, e_3, e_4), \quad e_k \in \mathcal{E},$$

with the ordered plaquette holonomy

$$U(P) := U(e_1)U(e_2)U(e_3)U(e_4);$$

3. the Wilson action

$$S_W(U) := \beta \sum_{P \in \mathcal{C}_\square} \left(1 - \frac{1}{3} \operatorname{Re} \operatorname{tr} U(P) \right), \quad \beta := \frac{6}{g_3^2};$$

4. the matter actions

$$S_F(U, \bar{\psi}, \psi) := \sum_{e \in \mathcal{E}} \bar{\psi}_{t(e)} (\psi_{t(e)} - U(e) \psi_{s(e)}) + m_F \sum_{x \in \mathcal{N}} \bar{\psi}_x \psi_x,$$

$$S_\phi(\phi) := \frac{1}{2} \sum_{e=(x \rightarrow y) \in \mathcal{E}} \frac{(\phi_y - \phi_x)^2}{\ell_e + \eta} + \sum_{x \in \mathcal{N}} \left(\frac{m_\phi^2}{2} \phi_x^2 + V_\phi(\phi_x) \right),$$

where ℓ_e is the spacetime length of edge e and $\eta > 0$ is the distance regularization from [2].

Remark 2.2. The actions above are identical in form to those in [2]; the only change is interpretive. In the companion paper the link variables were constructed from the particle data. In the present paper they are promoted to independent coordinates on $SU(3)^\mathcal{E}$.

2.2 Field manifold

Definition 2.3 (Gauge-link manifold). *Let*

$$\mathcal{U} := SU(3)^\mathcal{E} = \{U : \mathcal{E} \rightarrow SU(3)\}.$$

Because $\mathcal{E}$ is finite and $SU(3)$ is a compact smooth Lie group of real dimension 8, $\mathcal{U}$ is a compact smooth manifold of real dimension $8|\mathcal{E}|$.

Definition 2.4 (Auxiliary color manifold). *Let*

$$\mathbb{B} := \{z \in \mathbb{C}^3 : \|z\| < 1\}, \quad \mathcal{C} := \mathbb{B}^\mathcal{N}.$$

This is the product of the open unit balls suggested by the color normalization bound $\|\mathbf{c}_x\| < 1$ proved in [2]. Since no independent color action was supplied in the companion paper, this sector will be assigned a normalized product measure and decouples from the dynamical integrals below.

Definition 2.5 (Matter field domains). *Let*

$$\mathcal{S} := \mathbb{R}^\mathcal{N}$$

be the scalar field space. For the fermionic sector, write

$$V_F := \bigoplus_{x \in \mathcal{N}} \mathbb{C}_x^3$$

and let Λ_F be the Grassmann algebra generated by

$$\{\psi_{x,\alpha}, \bar{\psi}_{x,\alpha} : x \in \mathcal{N}, \alpha = 1, 2, 3\}.$$

The full promoted field manifold is

$$\mathcal{X}_\mathcal{L} := \mathcal{U} \times \mathcal{C} \times \Lambda_F \times \mathcal{S}.$$

Remark 2.6. Only the gauge and scalar sectors are manifolds in the ordinary sense. The fermionic sector is finite-dimensional Grassmann algebra, so integration in those variables is algebraic rather than measure theoretic. This is standard for finite lattice fermions [3, 4].

2.3 Measures

Definition 2.7 (Normalized Haar measure). *Let μ_H denote the unique normalized Haar probability measure on $SU(3)$, so that*

$$\mu_H(SU(3)) = 1$$

and for every Borel set $A \subseteq SU(3)$ and every $g \in SU(3)$,

$$\mu_H(gA) = \mu_H(Ag) = \mu_H(A).$$

The gauge-link measure on $\mathcal{U}$ is the product probability measure

$$d\mu_{\mathcal{U}}(U) := \prod_{e \in \mathcal{E}} d\mu_H(U(e)).$$

Definition 2.8 (Auxiliary color measure). *Let $\nu_{\mathbb{B}}$ be normalized Lebesgue measure on $\mathbb{B} \subset \mathbb{C}^3 \cong \mathbb{R}^6$. The color measure on $\mathcal{C}$ is*

$$d\nu_{\mathcal{C}}(c) := \prod_{x \in \mathcal{N}} d\nu_{\mathbb{B}}(c_x).$$

By normalization, $\nu_{\mathcal{C}}(\mathcal{C}) = 1$.

Definition 2.9 (Berezin and scalar measures). *The Berezin measure is*

$$d\mu_F := \prod_{x \in \mathcal{N}} \prod_{\alpha=1}^3 d\bar{\psi}_{x,\alpha} d\psi_{x,\alpha},$$

with the convention

$$\int d\bar{\psi} d\psi e^{-\bar{\psi}\psi} = 1.$$

The scalar measure on $\mathcal{S}$ is ordinary Lebesgue measure

$$d\phi := \prod_{x \in \mathcal{N}} d\phi_x.$$

Definition 2.10 (Total finite-lattice measure). *The total measure symbol on $\mathcal{X}_{\mathcal{L}}$ is*

$$d\mu := d\mu_{\mathcal{U}}(U) d\nu_{\mathcal{C}}(c) d\mu_F d\phi.$$

When the integrand does not depend on c , the auxiliary color sector contributes the unit factor

$$\int_{\mathcal{C}} d\nu_{\mathcal{C}}(c) = 1.$$

Proposition 2.11 (Well-posedness of the finite measure factors). *Each factor in Definition 2.10 is well defined. In particular, $\mu_{\mathcal{U}}$ is a Borel probability measure on a compact manifold, $\nu_{\mathcal{C}}$ is a Borel probability measure on a bounded open subset of $\mathbb{R}^{6|\mathcal{N}|}$, μ_F is a finite Berezin integration rule on a finite Grassmann algebra, and $\phi \mapsto S_{\phi}(\phi)$ is a measurable function on $\mathbb{R}^{|\mathcal{N}|}$.*

Proof. The product group $SU(3)^{\mathcal{E}}$ is compact because $\mathcal{E}$ is finite and $SU(3)$ is compact, hence the product Haar measure exists. The set $\mathcal{C} = \mathbb{B}^{\mathcal{N}}$ is a bounded open subset of a finite-dimensional Euclidean space, so the normalized Lebesgue measure exists. The fermionic sector has finitely many generators, hence finite Berezin integration is an algebraic operation. Finally S_{ϕ} is a finite sum of continuous functions of ϕ , because each coefficient $(\ell_e + \eta)^{-1}$ is finite and positive. $\square$

3 Construction of the partition function

The total Euclidean action. The quantized action on $\mathcal{X}_{\mathcal{L}}$ is the same algebraic expression as in the companion lattice paper, but now interpreted on the promoted field space:

$$S(U, \bar{\psi}, \psi, \phi) := S_W(U) + S_F(U, \bar{\psi}, \psi) + S_\phi(\phi).$$

The auxiliary color variable $c \in \mathcal{C}$ does not enter this action and is therefore normalized away. No hidden continuum limit is taken at this stage. Every term is a finite sum over the fixed finite index sets $\mathcal{N}$, $\mathcal{E}$, and $\mathcal{C}_\square$.

The master partition function. The central object of the paper is the finite-lattice partition function

$$Z := \int_{\mathcal{X}_{\mathcal{L}}} \exp(-S(U, \bar{\psi}, \psi, \phi)) \, d\mu. \quad (1)$$

Because the color sector is normalized and decoupled,

$$Z = \int_{\mathcal{U} \times \Lambda_F \times \mathcal{S}} \exp(-S_W(U) - S_F(U, \bar{\psi}, \psi) - S_\phi(\phi)) \, d\mu_{\mathcal{U}}(U) \, d\mu_F \, d\phi.$$

Normalization and expectation values. For any bounded gauge-invariant observable $\mathcal{O}$ whose fermionic degree is even, the vacuum expectation is defined by

$$\langle \mathcal{O} \rangle := \frac{1}{Z} \int_{\mathcal{X}_{\mathcal{L}}} \mathcal{O}(U, c, \bar{\psi}, \psi, \phi) e^{-S(U, \bar{\psi}, \psi, \phi)} \, d\mu, \quad (2)$$

provided $Z \neq 0$. The nonvanishing of Z is model dependent and cannot be asserted for arbitrary finite complex fermion determinants. What can always be proved is finiteness of the numerator and denominator as complex numbers, which is sufficient for all formal manipulations before division by Z .

Theorem 3.1 (Existence and finiteness of the partition function). *Under the hypotheses of Theorem 1.1, the partition function (1) is a finite complex number. Moreover, for every bounded measurable observable $\mathcal{O}$ depending polynomially on the Grassmann variables, the unnormalized expectation*

$$\int_{\mathcal{X}_{\mathcal{L}}} \mathcal{O} e^{-S} \, d\mu$$

is finite.

Proof. Integrate the fermions first. Section 7 will show that

$$\int_{\Lambda_F} e^{-S_F(U, \bar{\psi}, \psi)} \, d\mu_F = \det M_F(U),$$

where $M_F(U)$ is a finite complex matrix depending continuously on U . Since $\mathcal{U}$ is compact, the function $U \mapsto \det M_F(U)$ is bounded:

$$\sup_{U \in \mathcal{U}} |\det M_F(U)| < \infty.$$

The Wilson term satisfies $S_W(U) \geq 0$ for every U , because each summand $1 - \frac{1}{3} \operatorname{Re} \operatorname{tr} U(P)$ is nonnegative. Hence

$$|e^{-S_W(U)}| \leq 1.$$

For the scalar sector,

$$S_\phi(\phi) \geq \frac{m_\phi^2}{2} \sum_{x \in \mathcal{N}} \phi_x^2 + \sum_{x \in \mathcal{N}} V_\phi(\phi_x) \geq \frac{m_\phi^2}{2} \sum_{x \in \mathcal{N}} \phi_x^2 - C_\phi |\mathcal{N}|,$$

because the edge term is nonnegative and $V_\phi \geq -C_\phi$ by assumption. Therefore

$$e^{-S_\phi(\phi)} \leq e^{C_\phi |\mathcal{N}|} \exp\left(-\frac{m_\phi^2}{2} \sum_{x \in \mathcal{N}} \phi_x^2\right),$$

and the right-hand side is integrable on $\mathbb{R}^{|\mathcal{N}|}$. Putting the bounds together,

$$|Z| \leq \sup_{U \in \mathcal{U}} |\det M_F(U)| \cdot e^{C_\phi |\mathcal{N}|} \int_{\mathbb{R}^{|\mathcal{N}|}} e^{-\frac{m_\phi^2}{2} \sum_x \phi_x^2} d\phi < \infty.$$

The same domination argument works after multiplying by any bounded observable that is polynomial in the finitely many Grassmann generators, because Berezin integration of such a polynomial again yields a bounded continuous function of U . $\square$

4 Gauge redundancy and orbit analysis

The gauge orbit problem. The finite lattice theory has a local gauge symmetry at every site. This means that the coordinates $(U, \psi, \bar{\psi}, c)$ on field space are not physical coordinates: many different tuples encode the same gauge state. The redundancy is continuous, because the gauge group is a product of copies of the compact Lie group $SU(3)$.

Definition 4.1 (Local gauge group). *The lattice gauge group is*

$$\mathcal{G} := SU(3)^{\mathcal{N}} = \{\Omega : \mathcal{N} \rightarrow SU(3)\}.$$

Its action on the promoted fields is

$$U^\Omega(e) := \Omega(t(e)) U(e) \Omega(s(e))^{-1},$$

$$\psi_x^\Omega := \Omega(x) \psi_x, \quad \bar{\psi}_x^\Omega := \bar{\psi}_x \Omega(x)^{-1}, \quad c_x^\Omega := \Omega(x) c_x, \quad \phi_x^\Omega := \phi_x.$$

Proposition 4.2 (Gauge invariance of action and measure). *For every $\Omega \in \mathcal{G}$,*

$$S_W(U^\Omega) = S_W(U), \quad S_F(U^\Omega, \bar{\psi}^\Omega, \psi^\Omega) = S_F(U, \bar{\psi}, \psi), \quad S_\phi(\phi^\Omega) = S_\phi(\phi).$$

Moreover, $\mu_{\mathcal{U}}$, $\nu_{\mathcal{C}}$, and μ_F are invariant under the corresponding gauge transformations. Consequently the total weighted measure $e^{-S} d\mu$ is gauge invariant.

Proof. Gauge invariance of S_W and of the matter actions was already proved in [2]; the same algebra applies here because only the interpretation of the variables has changed. Haar invariance of $\mu_{\mathcal{U}}$ follows edgewise from left and right invariance of Haar measure on $SU(3)$. The normalized Lebesgue measure on the ball $\mathbb{B} \subset \mathbb{C}^3$ is invariant under the unitary action of $SU(3)$ because the action preserves Euclidean norms and Jacobian 1. Berezin measure is invariant under the linear change of variables $(\psi, \bar{\psi}) \mapsto (\Omega\psi, \bar{\psi}\Omega^{-1})$ because the Jacobians $\det \Omega$ and $\det \Omega^{-1}$ cancel and $\det \Omega = 1$ in $SU(3)$. $\square$

Definition 4.3 (Gauge orbit and stabilizer). *For a field configuration*

$$\Phi := (U, c, \bar{\psi}, \psi, \phi) \in \mathcal{X}_{\mathcal{L}}$$

its gauge orbit is

$$\mathcal{O}(\Phi) := \{\Phi^{\Omega} : \Omega \in \mathcal{G}\},$$

and its stabilizer is

$$\text{Stab}(\Phi) := \{\Omega \in \mathcal{G} : \Phi^{\Omega} = \Phi\}.$$

Symmetry redundancy versus physical degrees of freedom. The physical configuration space is not $\mathcal{X}_{\mathcal{L}}$ itself but the orbit space $\mathcal{X}_{\mathcal{L}}/\mathcal{G}$. On a finite graph one can count the redundant and nonredundant link degrees explicitly. This count is most transparent after choosing a spanning forest.

Definition 4.4 (Underlying undirected graph and component count). *Let $|\mathcal{L}|$ be the undirected graph obtained from $(\mathcal{N}, \mathcal{E})$ by forgetting orientation and identifying parallel opposite orientations when both are present. Let $b_0(|\mathcal{L}|)$ be the number of connected components of $|\mathcal{L}|$.*

Proposition 4.5 (Dimension count for gauge redundancy). *Assume the underlying undirected graph $|\mathcal{L}|$ has $b_0 := b_0(|\mathcal{L}|)$ connected components. Then:*

1. *the gauge-link manifold has real dimension $8|\mathcal{E}|$;*
2. *the rooted gauge group obtained by fixing one vertex per connected component has real dimension $8(|\mathcal{N}| - b_0)$;*
3. *after maximal-tree gauge fixing, the number of remaining independent gauge link coordinates is*

$$8(|\mathcal{E}| - |\mathcal{N}| + b_0),$$

which is 8 times the cycle rank of the graph.

Proof. Item (1) is immediate from $\dim SU(3) = 8$. For item (2), each unfixed site contributes one copy of $SU(3)$. For item (3), every spanning forest on a graph with $|\mathcal{N}|$ vertices and b_0 connected components has $|\mathcal{N}| - b_0$ edges, so removing exactly those tree-edge coordinates leaves $|\mathcal{E}| - |\mathcal{N}| + b_0$ cotree edges. Each cotree edge carries 8 real parameters. $\square$

Remark 4.6. The compactness of $SU(3)$ means that the gauge-group probability measure is finite when normalized. Thus on a finite lattice the problem with gauge redundancy is not an *infinite numerical volume divergence*; the problem is that the parametrization is many-to-one and therefore obscures the physical degrees of freedom. Gauge fixing removes this redundancy exactly.

5 Gauge fixing and the Faddeev–Popov procedure

Selecting the gauge condition. For a finite graph the cleanest globally valid gauge choice is a rooted maximal-tree gauge. Unlike lattice Landau gauge, it has no Gribov ambiguity: every rooted orbit intersects the gauge slice exactly once.

Definition 5.1 (Rooted spanning forest). *Choose one root vertex in each connected component of $|\mathcal{L}|$ and let $R \subseteq \mathcal{N}$ be the set of roots. A rooted spanning forest $\mathcal{T} \subseteq \mathcal{E}$ is an oriented set of edges such that:*

1. after forgetting orientation, $\mathcal{T}$ is a spanning forest of $|\mathcal{L}|$;
2. every edge of $\mathcal{T}$ is oriented away from its component root;
3. every nonroot vertex $x \in \mathcal{N} \setminus R$ has a unique parent edge

$$e_x = (p(x) \rightarrow x) \in \mathcal{T}.$$

Definition 5.2 (Rooted gauge group and maximal-tree gauge slice). *The rooted gauge group is*

$$\mathcal{G}_* := \{\Omega \in \mathcal{G} : \Omega(r) = I \text{ for every root } r \in R\}.$$

The maximal-tree gauge slice is

$$\Sigma_{\mathcal{T}} := \{U \in \mathcal{U} : U(e) = I \text{ for every } e \in \mathcal{T}\}.$$

Theorem 5.3 (Global rooted maximal-tree gauge). *For every $U \in \mathcal{U}$ there exists a unique $\Omega_U \in \mathcal{G}_*$ such that*

$$U^{\Omega_U} \in \Sigma_{\mathcal{T}}.$$

Equivalently, every rooted gauge orbit in $\mathcal{U}$ intersects $\Sigma_{\mathcal{T}}$ exactly once.

Proof. Existence is recursive along each tree component. Set $\Omega_U(r) = I$ at every root. Suppose $\Omega_U(p(x))$ is already defined for the parent of a nonroot vertex x . The condition

$$U^{\Omega_U}(e_x) = I$$

for the parent edge $e_x = (p(x) \rightarrow x)$ is

$$\Omega_U(x) U(e_x) \Omega_U(p(x))^{-1} = I,$$

hence

$$\Omega_U(x) = \Omega_U(p(x)) U(e_x)^{-1}.$$

Because every nonroot vertex has a unique parent and every component contains a unique path from the root to any vertex, this recursion defines Ω_U uniquely on all vertices. By construction $\Omega_U \in \mathcal{G}_*$ and $U^{\Omega_U}(e) = I$ for all $e \in \mathcal{T}$, so $U^{\Omega_U} \in \Sigma_{\mathcal{T}}$.

For uniqueness, let $\Omega_1, \Omega_2 \in \mathcal{G}_*$ satisfy $U^{\Omega_1}, U^{\Omega_2} \in \Sigma_{\mathcal{T}}$. Set $\Gamma := \Omega_2^{-1} \Omega_1 \in \mathcal{G}_*$. Then for every tree edge $e = (x \rightarrow y) \in \mathcal{T}$,

$$I = (U^{\Omega_1})(e) = \Gamma(y) (U^{\Omega_2})(e) \Gamma(x)^{-1} = \Gamma(y) \Gamma(x)^{-1}.$$

Thus $\Gamma(y) = \Gamma(x)$ on every tree edge. Since $\Gamma(r) = I$ on every root and every vertex is connected to a root by a unique tree path, induction along the tree gives $\Gamma(x) = I$ for all x . Hence $\Omega_1 = \Omega_2$. $\square$

The Faddeev–Popov identity. Because the gauge slice is global and unique, the Faddeev–Popov formula is particularly simple.

Definition 5.4 (Delta distribution on the tree gauge slice). *Let δ_I be the Dirac distribution at the identity of $SU(3)$, defined by*

$$\int_{SU(3)} f(V) \delta_I(V) d\mu_H(V) = f(I)$$

for continuous f . Define the tree-gauge delta distribution by

$$\delta_{\mathcal{T}}(U) := \prod_{e \in \mathcal{T}} \delta_I(U(e)).$$

Theorem 5.5 (Exact Faddeev–Popov identity in maximal-tree gauge). *For every $U \in \mathcal{U}$,*

$$1 = \int_{\mathcal{G}_*} \delta_{\mathcal{T}}(U^\Omega) \, d\mu_{\mathcal{G}_*}(\Omega),$$

where $\mu_{\mathcal{G}_*}$ is product normalized Haar measure on $\mathcal{G}_*$. Hence for every gauge-invariant integrable functional F on $\mathcal{U}$,

$$\int_{\mathcal{U}} F(U) \, d\mu_{\mathcal{U}}(U) = \int_{\mathcal{U}} F(U) \delta_{\mathcal{T}}(U) \, d\mu_{\mathcal{U}}(U).$$

Proof. Fix $U \in \mathcal{U}$. By Theorem 5.3, there is exactly one $\Omega_U \in \mathcal{G}_*$ such that $U^{\Omega_U} \in \Sigma_{\mathcal{T}}$. Therefore the integrand $\delta_{\mathcal{T}}(U^\Omega)$ vanishes for all $\Omega \neq \Omega_U$ and contributes 1 at $\Omega = \Omega_U$ by definition of the delta distribution, so the integral equals 1.

For the second identity, insert the first into the integral of F :

$$\int_{\mathcal{U}} F(U) \, d\mu_{\mathcal{U}}(U) = \int_{\mathcal{U}} \int_{\mathcal{G}_*} F(U) \delta_{\mathcal{T}}(U^\Omega) \, d\mu_{\mathcal{G}_*}(\Omega) \, d\mu_{\mathcal{U}}(U).$$

Use Fubini's theorem and then change variables $U \mapsto U^{\Omega^{-1}}$. Haar invariance implies $d\mu_{\mathcal{U}}(U^{\Omega^{-1}}) = d\mu_{\mathcal{U}}(U)$, and gauge invariance of F gives $F(U^{\Omega^{-1}}) = F(U)$. Thus

$$\int_{\mathcal{U}} F(U) \, d\mu_{\mathcal{U}}(U) = \int_{\mathcal{U}} F(U) \delta_{\mathcal{T}}(U) \, d\mu_{\mathcal{U}}(U) \int_{\mathcal{G}_*} d\mu_{\mathcal{G}_*}(\Omega).$$

The last factor equals 1 because $\mu_{\mathcal{G}_*}$ is normalized. □

The Faddeev–Popov determinant. The determinant can be computed directly from infinitesimal gauge variations. In maximal-tree gauge it is constant.

Definition 5.6 (Infinitesimal gauge parameters). *Let*

$$T^a := \frac{\lambda_a}{2}, \quad a = 1, \dots, 8,$$

where λ_a are the Gell-Mann matrices. For each site $x \in \mathcal{N}$ let

$$\omega_x = \sum_{a=1}^8 \omega_x^a T^a \in \mathfrak{h}_3,$$

and define the infinitesimal rooted gauge transformation

$$\Omega_\varepsilon(x) := \exp(i\varepsilon \omega_x), \quad \omega_r = 0 \text{ for every } r \in R.$$

Proposition 5.7 (Linearized gauge map on the tree). *Let $U \in \Sigma_{\mathcal{T}}$, so $U(e) = I$ for all $e \in \mathcal{T}$. For a tree edge $e = (x \rightarrow y) \in \mathcal{T}$,*

$$\left. \frac{d}{d\varepsilon} \right|_{\varepsilon=0} U^{\Omega_\varepsilon}(e) = i(\omega_y - \omega_x).$$

In the basis $\{T^a\}_{a=1}^8$, the linearized Faddeev–Popov operator is

$$M_{\text{FP}}^{(\mathcal{T})} = B_{\mathcal{T}} \otimes I_8,$$

where $B_{\mathcal{T}}$ is the rooted incidence matrix of the spanning forest.

Proof. Differentiate

$$U^{\Omega_\varepsilon}(e) = \Omega_\varepsilon(y) U(e) \Omega_\varepsilon(x)^{-1}$$

at $\varepsilon = 0$. Since $U(e) = I$ on the slice,

$$\left. \frac{d}{d\varepsilon} \right|_{\varepsilon=0} U^{\Omega_\varepsilon}(e) = i\omega_y - i\omega_x.$$

Writing each ω_x in the Gell-Mann basis shows that the eight color components decouple and each is acted on by the same incidence matrix $B_{\mathcal{T}}$. $\square$

Proposition 5.8 (The Faddeev–Popov determinant is identically one). *With the rooted maximal-tree gauge choice,*

$$\Delta_{\text{FP}}^{(\mathcal{T})} = \det M_{\text{FP}}^{(\mathcal{T})} = 1.$$

Proof. Order the nonroot vertices so that every parent precedes every child. Index the rows of $B_{\mathcal{T}}$ by the tree edges $e_x = (p(x) \rightarrow x)$ and the columns by the nonroot vertices x . Then the row corresponding to e_x has entry +1 in column x , entry -1 in column $p(x)$ if $p(x)$ is nonroot, and zero in all columns associated with vertices not on the path from the root to x . By the chosen ordering this matrix is triangular with diagonal entries all equal to 1. Hence

$$\det B_{\mathcal{T}} = 1.$$

Therefore

$$\det M_{\text{FP}}^{(\mathcal{T})} = \det(B_{\mathcal{T}} \otimes I_8) = (\det B_{\mathcal{T}})^8 = 1.$$

$\square$

Corollary 5.9 (Gauge-fixed partition function). *The partition function may be written as*

$$Z = \int_{\mathcal{U} \times \Lambda_{\text{F}} \times \mathcal{S}} \delta_{\mathcal{T}}(U) e^{-S_{\text{W}}(U) - S_{\text{F}}(U, \bar{\psi}, \psi) - S_{\phi}(\phi)} d\mu_{\mathcal{U}}(U) d\mu_{\text{F}} d\phi.$$

After integrating out the constrained tree links, this becomes an ordinary integral over the cotree edge variables

$$\mathcal{E}_{\text{co}} := \mathcal{E} \setminus \mathcal{T}.$$

Proof. Apply Theorem 5.5 to the gauge-invariant integrand obtained after combining the matter variables with Proposition 4.2. Since $\delta_{\mathcal{T}}(U)$ forces $U(e) = I$ on all tree edges, integrating over those edges contributes the unit factor

$$\int_{SU(3)} \delta_I(V) d\mu_H(V) = 1$$

edge by edge. $\square$

6 Ghost fields and the effective action

Introducing ghost fields. Although the Faddeev–Popov determinant is trivial in rooted maximal-tree gauge, it is still useful to represent it as a finite Berezin integral, because this makes the gauge-fixing algebra parallel to the standard continuum formalism.

Definition 6.1 (Ghost and antighost fields). *For each nonroot vertex $x \in \mathcal{N} \setminus R$, let*

$$c_x = \sum_{a=1}^8 c_x^a T^a$$

be an anticommuting $\mathfrak{h}_3$ -valued ghost field. For each tree edge $e \in \mathcal{T}$, let

$$\bar{c}_e = \sum_{a=1}^8 \bar{c}_e^a T^a$$

be an anticommuting $\mathfrak{h}_3$ -valued antighost field. Their Berezin measure is

$$d\mu_{\text{gh}} := \prod_{e \in \mathcal{T}} \prod_{a=1}^8 d\bar{c}_e^a \prod_{x \in \mathcal{N} \setminus R} \prod_{a=1}^8 dc_x^a.$$

Definition 6.2 (Ghost action in maximal-tree gauge). *Define*

$$S_{\text{ghost}}^{(\mathcal{T})}(\bar{c}, c) := \sum_{e=(x \rightarrow y) \in \mathcal{T}} \text{tr}(\bar{c}_e(c_y - c_x)),$$

with the convention $c_r := 0$ for root vertices $r \in R$.

Proposition 6.3 (Ghost representation of the Faddeev–Popov determinant). *The ghost integral reproduces the Faddeev–Popov determinant:*

$$\int e^{-S_{\text{ghost}}^{(\mathcal{T})}(\bar{c}, c)} d\mu_{\text{gh}} = \Delta_{\text{FP}}^{(\mathcal{T})} = 1.$$

Proof. Expand the ghost fields in the basis $\{T^a\}_{a=1}^8$. Then the action factorizes over color indices:

$$S_{\text{ghost}}^{(\mathcal{T})}(\bar{c}, c) = \sum_{a=1}^8 \sum_{e=(x \rightarrow y) \in \mathcal{T}} \bar{c}_e^a (c_y^a - c_x^a).$$

This is precisely the quadratic Berezin form associated with the matrix $B_{\mathcal{T}} \otimes I_8$, so the standard finite-dimensional Berezin identity gives

$$\int e^{-S_{\text{ghost}}^{(\mathcal{T})}} d\mu_{\text{gh}} = \det(B_{\mathcal{T}} \otimes I_8).$$

Proposition 5.8 gives the value 1. □

The effective quantized action. To display the standard BRST structure it is convenient to introduce auxiliary bosonic Nakanishi–Lautrup fields $b_e \in \mathfrak{h}_3$ for $e \in \mathcal{T}$.

Definition 6.4 (BRST differential and gauge-fixing fermion). *Let s act on the full field algebra by*

$$\begin{aligned} sU(e) &= ic_{t(e)}U(e) - iU(e)c_{s(e)}, & s\psi_x &= ic_x\psi_x, & s\bar{\psi}_x &= -i\bar{\psi}_xc_x, \\ s\phi_x &= 0, & sc_x &= -ic_x^2, & s\bar{c}_e &= b_e, & sb_e &= 0. \end{aligned}$$

Define the gauge-fixing fermion

$$\Psi_{\mathcal{T}} := \sum_{e \in \mathcal{T}} \text{tr} \left(\bar{c}_e \left(U(e) - I + \frac{\alpha}{2} b_e \right) \right), \quad \alpha \geq 0.$$

Lemma 6.5 (Nilpotency of the BRST differential). *The operator s satisfies $s^2 = 0$ on generators and therefore on the whole field algebra.*

Proof. For ϕ_x and b_e the claim is immediate. For $\bar{c}_e$, $s^2\bar{c}_e = sb_e = 0$. For c_x ,

$$s^2c_x = -i s(c_x^2) = -i((sc_x)c_x - c_x(sc_x)) = -i((-ic_x^2)c_x - c_x(-ic_x^2)) = 0.$$

For ψ_x ,

$$s^2\psi_x = i(sc_x)\psi_x - ic_x(s\psi_x) = i(-ic_x^2)\psi_x - ic_x(ic_x\psi_x) = 0,$$

and similarly for $\bar{\psi}_x$. Finally, for a link variable $U(e)$,

$$\begin{aligned} s^2U(e) &= i(sc_{t(e)})U(e) + ic_{t(e)}sU(e) - isU(e)c_{s(e)} - iU(e)sc_{s(e)} \\ &= c_{t(e)}^2U(e) - c_{t(e)}^2U(e) + U(e)c_{s(e)}^2 - U(e)c_{s(e)}^2 = 0. \end{aligned}$$

□

Proposition 6.6 (BRST-exact gauge fixing). *The gauge-fixing and ghost terms are*

$$S_{\text{gf+gh}}^{(\mathcal{T})} := s\Psi_{\mathcal{T}}.$$

Explicitly,

$$S_{\text{gf+gh}}^{(\mathcal{T})} = \sum_{e \in \mathcal{T}} \text{tr} \left(b_e(U(e) - I) + \frac{\alpha}{2} b_e^2 \bar{c}_e (c_{t(e)}U(e) - U(e)c_{s(e)}) \right).$$

Consequently the effective action

$$S_{\text{eff}} := S_W + S_F + S_\phi + S_{\text{gf+gh}}^{(\mathcal{T})}$$

is BRST invariant:

$$sS_{\text{eff}} = 0.$$

Proof. Apply s to $\Psi_{\mathcal{T}}$ and use Definition 6.4. Since the fields b_e commute with everything and the ghosts anticommute, one obtains the displayed formula. Gauge invariance of $S_W + S_F + S_\phi$ implies $s(S_W + S_F + S_\phi) = 0$, because s is the infinitesimal gauge differential with ghost parameter c . Then

$$sS_{\text{eff}} = s(S_W + S_F + S_\phi) + s^2\Psi_{\mathcal{T}} = 0$$

by Lemma 6.5. □

Remark 6.7. At $\alpha = 0$ the b -integration imposes the sharp gauge condition $U(e) = I$ on tree edges. Since the Faddeev–Popov determinant is constant, the ghost sector decouples after gauge fixing. The BRST presentation is therefore exact but algebraically simpler than in continuum gauges where the determinant depends on the gauge field.

7 Matter sector integration

Fermionic Berezin integration. On a finite lattice the fermion integral is exact and elementary once the quadratic form is written in matrix notation.

Definition 7.1 (Fermion matrix). *Order the site-color basis vectors by pairs (x, α) with $x \in \mathcal{N}$ and $\alpha \in \{1, 2, 3\}$. Define the matrix $M_F(U) \in \mathbb{C}^{3|\mathcal{N}| \times 3|\mathcal{N}|}$ by*

$$(M_F(U))_{(y,\alpha),(x,\beta)} := (m_F + \text{indeg}(y)) \delta_{xy} \delta_{\alpha\beta} - \sum_{\substack{e \in \mathcal{E} \\ s(e)=x, t(e)=y}} U(e)_{\alpha\beta},$$

where

$$\text{indeg}(y) := \#\{e \in \mathcal{E} : t(e) = y\}.$$

Lemma 7.2 (Quadratic form identity). *With $M_F(U)$ from Definition 7.1,*

$$S_F(U, \bar{\psi}, \psi) = \sum_{x,y \in \mathcal{N}} \sum_{\alpha,\beta=1}^3 \bar{\psi}_{y,\alpha} (M_F(U))_{(y,\alpha),(x,\beta)} \psi_{x,\beta}.$$

Proof. Expand the edge sum in Definition 2.1:

$$\sum_{e \in \mathcal{E}} \bar{\psi}_{t(e)} \psi_{t(e)} = \sum_{y \in \mathcal{N}} \text{indeg}(y) \bar{\psi}_y \psi_y.$$

The remaining transported terms contribute the off-diagonal blocks $-\sum_{e:s(e)=x,t(e)=y} U(e)$. Adding the mass term gives the stated matrix. $\square$

Theorem 7.3 (Fermion determinant). *For every gauge field $U \in \mathcal{U}$,*

$$\int_{\Lambda_F} e^{-S_F(U, \bar{\psi}, \psi)} d\mu_F = \det M_F(U).$$

Consequently

$$Z = \int_{\mathcal{U} \times \mathcal{S}} e^{-S_W(U) - S_\phi(\phi)} \det M_F(U) d\mu_{\mathcal{U}}(U) d\phi.$$

Proof. By Lemma 7.2, the exponent is the standard quadratic Berezin form $\bar{\Psi} M_F(U) \Psi$ in finitely many Grassmann variables. The finite-dimensional Berezin integral identity therefore gives

$$\int e^{-\bar{\Psi} M \Psi} d\bar{\Psi} d\Psi = \det M$$

for any finite matrix M . Applying this with $M = M_F(U)$ yields the first claim. Substituting the result into the definition of Z gives the second claim. $\square$

Remark 7.4 (Virtual loops and graph support). The determinant $\det M_F(U)$ is a finite polynomial in the link matrices and encodes the usual closed virtual-fermion loops. Because the link set $\mathcal{E}$ consists of the worldline and interaction edges from the trajectory data, only loops in that explicit graph contribute. The interaction-graph weights $\omega_{ij}^{\text{link}}(t)$ from [2] influence these virtual loops through the dependence of the original data-induced background links on the viscous force and its weights.

Scalar kinetic fluctuations. The scalar action is purely bosonic, real valued, and bounded below as soon as the local potential is bounded below. The regularization $\eta > 0$ excludes edge-length singularities.

Definition 7.5 (Scalar stiffness operator). *Define the symmetric quadratic form*

$$Q_\eta(\phi) := \frac{1}{2} \sum_{e=(x \rightarrow y) \in \mathcal{E}} \frac{(\phi_y - \phi_x)^2}{\ell_e + \eta}.$$

There is a unique symmetric matrix $L_\eta \in \mathbb{R}^{|\mathcal{N}| \times |\mathcal{N}|}$ such that

$$Q_\eta(\phi) = \frac{1}{2} \phi^\top L_\eta \phi.$$

Lemma 7.6 (Positivity and bounded coefficients). *The operator L_η is positive semidefinite. Moreover every edge weight*

$$w_e := \frac{1}{\ell_e + \eta}$$

satisfies

$$0 < w_e \leq \eta^{-1}.$$

Proof. Each summand in $Q_\eta(\phi)$ is nonnegative, so $Q_\eta(\phi) \geq 0$ for every ϕ , hence L_η is positive semidefinite. Since $\ell_e \geq 0$ and $\eta > 0$,

$$\ell_e + \eta \geq \eta,$$

which implies the upper bound on w_e . □

Theorem 7.7 (Scalar integrability). *Assume V_ϕ is continuous and bounded below by $-C_\phi$. Then*

$$Z_\phi := \int_{\mathcal{S}} e^{-S_\phi(\phi)} d\phi$$

is finite. If in addition $V_\phi \equiv 0$, then

$$Z_\phi = (2\pi)^{|\mathcal{N}|/2} \det(L_\eta + m_\phi^2 I)^{-1/2}.$$

Proof. By Definition 7.5,

$$S_\phi(\phi) = \frac{1}{2} \phi^\top L_\eta \phi + \frac{m_\phi^2}{2} \phi^\top \phi + \sum_{x \in \mathcal{N}} V_\phi(\phi_x).$$

Lemma 7.6 implies the first quadratic term is nonnegative, while the lower bound on V_ϕ gives

$$S_\phi(\phi) \geq \frac{m_\phi^2}{2} \phi^\top \phi - C_\phi |\mathcal{N}|.$$

Thus

$$e^{-S_\phi(\phi)} \leq e^{C_\phi |\mathcal{N}|} e^{-\frac{1}{2} m_\phi^2 \phi^\top \phi},$$

and the right-hand side is integrable on $\mathbb{R}^{|\mathcal{N}|}$.

If $V_\phi \equiv 0$, then

$$S_\phi(\phi) = \frac{1}{2} \phi^\top (L_\eta + m_\phi^2 I) \phi.$$

Since L_η is positive semidefinite and $m_\phi^2 I$ is strictly positive, the matrix $L_\eta + m_\phi^2 I$ is positive definite. The standard Gaussian integral formula gives the displayed determinant expression. □

Remark 7.8 (Role of η). The parameter $\eta > 0$ is not a perturbative convenience; it is the exact mechanism that prevents the scalar kinetic coefficients from diverging when two sites become spacetime-coincident. Without η , the coefficient $1/\ell_e$ would be singular on zero-length edges. With $\eta > 0$, every kinetic coefficient is uniformly bounded by η^{-1} .

8 Recovery of the Feynman path-integral measure

The continuum limit of the measure. The companion lattice paper proves that, under an admissible mean-field refinement regime, the Wilson action converges to the continuum Yang–Mills action on smooth fields [2]. To understand the measure side of the formal continuum path integral one must compare Haar measure on $SU(3)$ with Lebesgue measure on the Lie algebra near the identity.

Definition 8.1 (Lie-algebra coordinates). *Let*

$$\mathfrak{su}(3) = i\mathfrak{h}_3$$

be the Lie algebra of $SU(3)$. Equip $\mathfrak{h}_3$ with the Euclidean inner product

$$\langle H, K \rangle := \frac{1}{2} \operatorname{tr}(HK).$$

Choose an open neighborhood $\mathcal{V} \subset \mathfrak{h}_3$ of 0 on which the map

$$H \mapsto \exp(iH)$$

is a diffeomorphism onto an open neighborhood $\mathcal{U} \subset SU(3)$ of the identity.

Proposition 8.2 (Single-edge Haar density in exponential coordinates). *There exists a smooth strictly positive function*

$$J : \mathcal{V} \rightarrow (0, \infty)$$

with $J(0) = 1$ such that for every continuous function f supported in $\mathcal{U}$,

$$\int_{SU(3)} f(U) \, d\mu_H(U) = \int_{\mathcal{V}} f(e^{iH}) J(H) \, dH,$$

where dH is Lebesgue measure on the 8-dimensional Euclidean space $\mathfrak{h}_3$. Moreover,

$$J(H) = 1 + O(\|H\|^2) \quad \text{as } H \rightarrow 0.$$

Proof. The push-forward of μ_H by the chart inverse $U \mapsto H$ is a smooth measure on the Euclidean chart $\mathcal{V}$ and therefore has a smooth density $J(H)$ with respect to Lebesgue measure. The density is strictly positive because Haar measure has full support and the chart is a diffeomorphism. By rescaling the chosen Lebesgue measure on $\mathfrak{h}_3$ if necessary, one may arrange $J(0) = 1$. Since J is smooth and the Haar measure is inversion invariant, the linear term in the Taylor expansion vanishes at the identity, so $J(H) = 1 + O(\|H\|^2)$. $\square$

Corollary 8.3 (Scaled near-identity measure). *Fix $A \in \mathfrak{h}_3$ and write*

$$U_a := \exp(iaA)$$

for $a > 0$ small enough. Then

$$d\mu_H(U_a) = a^8 (1 + O(a^2)) \, dA$$

in exponential coordinates.

Proof. Apply Proposition 8.2 with $H = aA$. Since $dH = a^8 dA$ in an 8-dimensional space and $J(aA) = 1 + O(a^2)$, the formula follows. $\square$

Emergent regularization. The finite lattice is an ultraviolet regularization in the strict, literal sense that there are only finitely many integration variables and every coefficient in the action is finite. No ultraviolet divergence can appear before one takes a refinement limit.

Proposition 8.4 (Finite lattice spacing as an ultraviolet cutoff). *On any fixed data set $\mathcal{Q}$, the quantized theory has finitely many gauge variables, finitely many scalar variables, and finitely many fermionic generators. Consequently all unnormalized correlation functions are finite numbers. Any ultraviolet divergence can arise only from a limit in which the number of lattice cells tends to infinity and the mesh size tends to zero.*

Proof. The set $\mathcal{E}$ is finite by construction in [2], hence the gauge sector is the compact manifold $SU(3)^{\mathcal{E}}$. The scalar sector is $\mathbb{R}^{|\mathcal{N}|}$ with $|\mathcal{N}| < \infty$, and the fermionic sector has finitely many Grassmann generators. By Theorem 3.1, all unnormalized correlation functions are finite on the fixed lattice. Therefore no divergence is present at finite mesh. Such divergences can only appear in a subsequent limit with increasing numbers of lattice degrees of freedom. $\square$

Consistency with continuum QFT. The correct honest statement is not that we have already constructed the full continuum quantum measure, but that the local building blocks of the discrete measure and action match the formal continuum Feynman rules in the small-mesh, near-identity regime.

Theorem 8.5 (Local consistency with the formal continuum Yang–Mills measure). *Assume the admissible mean-field refinement regime of [2] and the continuum Wilson-action limit proved there. Let F be a bounded continuous function of finitely many Lie-algebra coordinates $A_{e_1}, \dots, A_{e_m}$, and for each refinement level N write*

$$U^{(N)}(e_k) = \exp(ia_N A_{e_k})$$

on the edges on which F depends. Then the corresponding cylinder-measure element satisfies

$$\prod_{k=1}^m d\mu_H(U^{(N)}(e_k)) = a_N^{8m} (1 + O(a_N^2)) \prod_{k=1}^m dA_{e_k},$$

and the weighted cylinder integrand has the same leading form as the formal continuum Euclidean Yang–Mills weight

$$\exp\left(-\frac{1}{4} \int_M \text{tr}(F_{\mu\nu} F^{\mu\nu}) \, \text{dvol}_g\right) \mathcal{D}A$$

when tested against the finite-coordinate observable F .

Proof. The measure statement is just the m -fold product of Corollary 8.3. The action statement follows from the continuum limit theorem proved in [2], which identifies the leading Wilson action density with the continuum Yang–Mills density. Combining the two gives the claimed leading cylinder-weight formula. $\square$

Remark 8.6. Theorem 8.5 is a consistency theorem for local charts and cylinder observables, not a constructive theorem producing a continuum probability measure on the infinite-dimensional space of gauge fields. A full continuum construction would require tightness, compatibility of cylinder measures, reflection positivity, and nontrivial control of the interacting limit. Those steps are not proved in this paper.

9 Discussion of excitations and propagators

The propagator on the interaction graph. Because the theory is finite, all correlation functions can be defined by source differentiation exactly. Their graph-theoretic support is determined by the explicit trajectory-induced lattice.

Definition 9.1 (Generating functional). *Introduce scalar sources $J = (J_x)_{x \in \mathcal{N}}$ and fermionic sources $\eta = (\eta_x)_{x \in \mathcal{N}}$, $\bar{\eta} = (\bar{\eta}_x)_{x \in \mathcal{N}}$. The generating functional is*

$$Z[J, \eta, \bar{\eta}] := \int \exp \left(-S + \sum_{x \in \mathcal{N}} J_x \phi_x + \sum_{x \in \mathcal{N}} \bar{\eta}_x \psi_x + \sum_{x \in \mathcal{N}} \bar{\psi}_x \eta_x \right) d\mu.$$

Definition 9.2 (Two-point functions). *Whenever $Z[0, 0, 0] \neq 0$, the scalar and fermion two-point functions are*

$$G_\phi(x, y) := \langle \phi_x \phi_y \rangle, \quad S_F(x, y)_{\alpha\beta} := \langle \psi_{x,\alpha} \bar{\psi}_{y,\beta} \rangle.$$

Proposition 9.3 (Background-field propagators). *Fix a gauge field U .*

1. *If $V_\phi \equiv 0$, then the scalar background covariance is*

$$G_\phi^{(U)} = (L_\eta + m_\phi^2 I)^{-1}.$$

2. *If $M_F(U)$ is invertible, then the fermion background covariance is*

$$S_F^{(U)} = M_F(U)^{-1}.$$

Proof. The scalar claim is the standard Gaussian covariance formula from Theorem 7.7. The fermion claim follows from differentiating the Berezin generating functional

$$\int e^{-\bar{\Psi} M_F(U) \Psi + \bar{\eta} \Psi + \bar{\Psi} \eta} d\bar{\Psi} d\Psi = \det M_F(U) e^{\bar{\eta} M_F(U)^{-1} \eta}.$$

□

Proposition 9.4 (Large-mass hopping expansion). *Assume $M_F(U) = D - H(U)$ where*

$$D := \text{diag}(m_F + \text{indeg}(x))_{x \in \mathcal{N}} \otimes I_3$$

and suppose

$$\|D^{-1}H(U)\| < 1$$

in operator norm. Then

$$M_F(U)^{-1} = \sum_{n=0}^{\infty} (D^{-1}H(U))^n D^{-1},$$

and the (x, y) matrix element is a convergent sum over directed paths $\gamma : y \rightsquigarrow x$ in $\mathcal{E}$. Therefore the support of the fermion propagator is controlled by the explicit interaction/worldline graph, and the same-time path combinatorics are inherited from the interaction data $\mathcal{P}_t$ and hence from the stochastic weights $\omega_{ij}^{\text{link}}(t)$ used in [2].

Proof. The Neumann series for $(I - D^{-1}H)^{-1}$ converges in operator norm under the assumed bound. Each power $(D^{-1}H)^n$ is a sum over length- n compositions of off-diagonal hopping operators, hence over directed paths of length n in the graph. The only same-time edges available for those paths are the interaction edges determined by the data sets $\mathcal{P}_t$, which were constructed from the weights $\omega_{ij}^{\text{link}}(t)$. □

Wilson loops and the area law. The Wilson loop remains the basic gauge-invariant probe of confinement.

Definition 9.5 (Wilson loop expectation). *For an oriented closed loop γ in the lattice,*

$$W(\gamma) := \text{tr} \left(\prod_{e \in \gamma} U(e) \right), \quad \langle W(\gamma) \rangle := Z^{-1} \int W(\gamma) e^{-S} d\mu,$$

whenever $Z \neq 0$.

Remark 9.6 (Area-law criterion). If a family of loops γ admits spanning surfaces Σ_γ in the embedded spacetime geometry and one can prove

$$|\langle W(\gamma) \rangle| \leq C e^{-\sigma \text{Area}(\Sigma_\gamma)}$$

with $\sigma > 0$ independent of the loop size, then one has the usual Euclidean area-law signature of confinement [5]. The present paper does *not* establish such a bound; it merely provides the exact finite-lattice measure in which the question is well posed.

10 Conclusion

Summary of the quantum data package. Starting from the finite lattice data package $\mathcal{Q}$ built in [2], this paper has completed the exact finite-dimensional quantization step. The link variables were promoted from data-derived transporters to independent $SU(3)$ variables, the full configuration space and product measure were defined, and the partition function was proved finite under the minimal scalar lower-bound assumption needed for scalar integrability. The gauge redundancy was solved globally by a rooted maximal-tree gauge, with exact orbit uniqueness, exact Faddeev–Popov identity, constant determinant, and a decoupled ghost sector. The fermions integrate to a finite determinant, the scalar sector is controlled by a positive stiffness operator regularized by η , and the local measure/action asymptotics match the standard formal Euclidean Yang–Mills rules in the smooth small-mesh regime.

Forward look to the mass-gap problem. This is the correct finite-lattice foundation for any later mass-gap analysis, but it is not itself a proof of a mass gap and not an unconditional construction of continuum quantum Yang–Mills theory. To move from the present paper to a Millennium-problem-level result, one would still need at least the following genuinely new ingredients: a nontrivial continuum limit of the gauge-fixed lattice measures; verification of reflection positivity and the full Euclidean axioms in that limit; reconstruction of a Hilbert-space quantum theory; and finally a proof that the energy–momentum spectrum has a strictly positive gap above the vacuum. Those problems remain open here. What this paper contributes is the rigorous finite-dimensional quantization step on the trajectory-induced lattice and the precise identification of the remaining gap between that step and a full constructive Yang–Mills existence theorem.

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